Lab Exercise 6: Simple Calculator App with AI Assistance

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Learning Outcomes

By completing this lab experiment, students will be able to:

- Use addEventListener() to handle click and mouse events.
- Retrieve user input from HTML input fields using JavaScript.
- Perform basic arithmetic operations using JavaScript.
- Display calculation results dynamically using textContent.
- Manipulate HTML elements using JavaScript DOM methods.
- Use CSS to style an interactive calculator interface.
- Use AI tools to assist with coding, debugging, and styling webpages.

Objectives

This lab experiment introduces students to event-driven programming through the development of a simple calculator.

Students will learn how to:

- Accept two numbers from the user.
- Perform addition, subtraction, multiplication, and division.

- Respond to button clicks.
- Change button appearance when the mouse moves over it.
- Display calculation results and error messages.
- Use AI assistance to improve and debug their code.

Students are encouraged to use AI tools to:

- Explain JavaScript event listeners.
- Debug arithmetic calculations.
- Improve calculator styling.
- Explain how to retrieve values from input fields.
- Add optional features to the calculator.

Lab Exercise

Instructions

- Complete the lab exercise.
- Save all work in separate HTML, CSS, and JavaScript files.

Description

Create a simple calculator webpage that allows users to enter two numbers and perform addition, subtraction, multiplication, and division.

The calculator should contain two input fields, four operation buttons, and a result area. When a user clicks an operation button, the calculator performs the selected operation and displays the result.

Additionally, when the mouse pointer moves over an operation button, the button's background color should change. When the mouse pointer leaves the button, its original color should be restored.

Requirements

HTML Requirements

- Create two input fields for entering numbers.

- Create four buttons labeled +, −, ×, and ÷.
- Create a result area using a <p> or element.
- Assign appropriate IDs to the input fields, buttons, and result area.

CSS Requirements

- Style the calculator using a centered layout.
- Set appropriate widths for the input fields and buttons.
- Use different background colors for the calculator buttons if desired.
- Add spacing between elements.
- Use CSS to create a visual hover effect for the operation buttons.

JavaScript Requirements

- Use document.queryselector() to select the HTML elements.
- Use addEventListener("click", ...) to handle the four operation buttons.
- Retrieve the two numbers using the value property.
- Convert input values to numbers before performing calculations.
- Display the result using textContent.
- Use addEventListener("mouseover", ...) to change a button's background color.
- Use addEventListener("mouseout", ...) to restore the button's original background color.
- Display an error message when the user attempts to divide by zero.
- Handle empty input fields appropriately.

Sample Output Behavior

User Action	Expected Result
Enter 10 and 5, then click +	Result: 15
Enter 10 and 5, then click −	Result: 5
Enter 10 and 5, then click ×	Result: 50
Enter 10 and 5, then click ÷	Result: 2
Move the mouse over an operation button	The button changes color
Move the mouse away from the button	The original button color returns
Divide by zero	An error message is displayed

Sample HTML Code

Save the following code as **index.html**.

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-
scale=1.0">
    <title>Simple Calculator</title>
    <link rel="stylesheet" href="style.css">
</head>
<body>

  <h1>Simple Calculator</h1>
  <div id="calculator">
     <input type="number" id="num1" placeholder="Enter first
number">
     <input type="number" id="num2" placeholder="Enter second
number">

     <div id="buttons">
        <button id="addBtn">+</button>
        <button id="subtractBtn">−</button>
        <button id="multiplyBtn">×</button>
        <button id="divideBtn">÷</button>
     </div>

     <p id="result">Result: </p>
    </div>

    <script src="script.js"></script>
</body>
</html>
```

Sample CSS Code

Save the following code as style.css.

```
body {
    font-family: Arial, sans-serif;
    text-align: center;
    background-color: #f4f7fb;
    margin-top: 50px;
}
```

```
#calculator {
    width: 320px;
    margin: auto;
    padding: 20px;
    background-color: white;
    border-radius: 12px;
    box-shadow: 0 2px 8px #cccccc;
}

input {
    width: 90%;
    padding: 10px;
    margin: 8px 0;
    box-sizing: border-box;
}

button {
    width: 55px;
    padding: 10px;
    margin: 5px;
    font-size: 1.2rem;
    background-color: steelblue;
    color: white;
    border: none;
    border-radius: 5px;
    cursor: pointer;
}

#result {
    font-size: 1.2rem;
    font-weight: bold;
    margin-top: 20px;
}
```

Sample JavaScript Code

Save the following code as script.js. Complete the missing sections.

```
/*
Name: John Smith
Course: JavaScript Programming
Lab Experiment: Simple Calculator and AI Assistance
Date:
```

```
*/

console.log("------ Lab Exercise 6: Simple Calculator ------");

// Collect the input fields and result element


// Collect the four operation buttons


// Create a function to perform a calculation
// 1. Check whether both input fields contain values.
// 2. Convert the values to numbers.
// 3. Perform the selected arithmetic operation.
// 4. Display the result using textContent.
// 5. Handle division by zero.


// Add click event listeners to the four buttons


// Add mouseover event listeners to change button colors


// Add mouseout event listeners to restore button colors



/*
AI Assistance:
Describe how AI helped you understand JavaScript events,
perform calculations, or debug your calculator.
*/
```

AI Assistance Activity

Instructions

Open an AI assistant and ask one or more of the following questions:

1. How does addEventListener("click", ...) work in JavaScript?
2. How can I retrieve a number from an HTML input field?
3. What is the difference between value and textContent?
4. How can I use one JavaScript function to perform four arithmetic operations?
5. How can I use mouseover and mouseout to change a button's background color?
6. How can I prevent division by zero and handle empty input fields?

After completing the lab, add a short comment using /* */ at the end of your .js file explaining how AI helped you during the activity. Include at least one example of a suggestion you used or evaluated.

Submission Requirements

- Upload your HTML, CSS, and JavaScript files to your GitHub repository.
- Test all four arithmetic operations and both mouse events.
- Verify that empty inputs and division by zero are handled appropriately.
- Submit the HTTPS link to your GitHub repository.

Grading Criteria

- Calculator functionality: HTML, CSS, and JavaScript: 7 pts
- AI assistance activity and reflection: 3 pts
- **Total: 10pts**